SUBHO SAHA

Electrical Engineering Student | Robotics & Embedded Systems Developer

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PROFESSIONAL SUMMARY

Motivated Electrical Engineering student at Kalyani Government Engineering College with a strong passion for robotics, embedded systems, IoT, and automation. Experienced in designing and developing autonomous drones, maze-solving robots, smart vehicles, and embedded control systems using ESP32 and Arduino platforms. Skilled in hardware-software integration, sensor fusion, control systems, and full-stack web technologies. Seeking opportunities to apply engineering knowledge to innovative real-world projects.

EDUCATION

Bachelor of Technology (B.Tech) in Electrical Engineering

Kalyani Government Engineering College (KGEC), MAKAUT
2024 – Present

- YGPA: 7.6 / 10

ACADEMIC ACHIEVEMENTS

- WBJEE 2024 – AIR 3513
- JEE Main 2024 – 96 Percentile
- Higher Secondary Examination – 94%

TECHNICAL SKILLS

Programming Languages

- C
- C++

- Python
- JavaScript
- HTML
- CSS
- Node.js

Embedded Systems & IoT

- ESP32
- Arduino
- ESP-NOW Communication
- Sensor Integration
- PWM Control
- PID Control
- I2C / SPI / UART Communication

Robotics & Automation

- Autonomous Drones
- Line Following Robots
- Maze Solving Robots
- 4WD Smart Vehicles
- Sensor Fusion
- Motion Control Systems

Simulation & Design Tools

- MATLAB
- Proteus
- Multisim
- TinkerCAD
- AutoCAD
- SolidWorks

Industrial & Electrical Tools

- PLC
- SCADA
- LabVIEW
- Power Systems

- Electrical Machines
- Battery Management Systems
- PCB Design Fundamentals

PROJECTS

ESP32-Based Autonomous Quadcopter Drone

- Designed and developed an autonomous quadcopter using ESP32 as the flight controller.
- Integrated MPU9250 IMU, ultrasonic sensors, and ToF sensors for stabilization and altitude estimation.
- Implemented sensor fusion, auto-leveling, failsafe landing, and ESP-NOW communication.
- Built on an F450 frame with DJI 2212 motors and 3S LiPo battery system.

4WD Smart Terrain Vehicle

- Developed a high-performance 4WD robotic vehicle capable of climbing steep inclines and navigating rough terrain.
- Implemented differential drive control and obstacle detection systems.
- Designed for sand, gravel, bridge, and water traversal challenges.

MAZEMERIZE Autonomous Maze Solver

- Designed and programmed an autonomous maze-solving robot.
- Implemented line-following algorithms and path optimization techniques.
- Integrated multiple sensors for accurate navigation and decision making.

Hope Drone Web Application

- Developed a web-based drone monitoring and control dashboard.
- Built using modern web technologies for telemetry visualization and system monitoring.

Line Following Robot

- Designed and developed a PID-controlled autonomous line-following robot.
- Optimized sensor calibration and motor control algorithms for high-speed operation.

LANGUAGES

- English (Fluent)
- Hindi (Fluent)
- Bengali (Native)

INTERESTS

- Robotics
- Embedded Systems
- Drone Technology
- IoT Development
- Automation Systems
- Power Systems Engineering
- Autonomous Vehicles

CAREER OBJECTIVE

To build innovative robotics, automation, and embedded systems solutions that solve real-world engineering challenges while continuously expanding my expertise in electrical engineering, intelligent systems, and advanced technology development.